

DivPredictionBot audit

Date: 2026-10-06 | Repo: C:\repos\prive\Bots\DivPredictionBot | Files: Worker.cs, McbProbabilityEngine.cs (plus CandlesBot, BotShared TimeframeBucketing/SymbolCandleAggregator and MyTradingBox mcb-prediction-lines.ts for context). No code was changed.

Summary: three real bugs (wrong live candle for higher timeframes, monthly treated as 1 minute, money-flow lines switching sides), two design problems (tick backlog, box writes every tick) and some smaller issues.

Recommended order: fix 1, 2 and 3 first (small fixes, direct cause of wrong predictions and misplaced lines), then 4 and 5. Open question for 5: should this bot own `boxes.position_type` for all boxes?

Bugs

1. The live candle for every timeframe above 1m is built from a single minute

Probably the main cause of the remaining 20%. CandlesBot only saves **closed** candles (`.Where(c => IsCandleClosed(...))` in `FetchOnlyNewCandles`), so the current 1h/4h/1d/1w bucket is never in the database. `Worker.cs:324-339` then always takes the "new bucket" branch and starts that candle from the current 1m candle alone.

- The live 1d candle gets the open, high and low of the current minute only. The merge branch above it (`Worker.cs:308-323`) never runs in practice.
- This puts `high[last]/low[last]`, `hlc3` (so the wave), ATR and the session's highest price on that bar wrong. If price made a higher high earlier in the hour and has since pulled back, the bear divergence isn't seen.
- For 6m/12m/24m the importer waits 60 s after a bucket closes, so during that first minute the previous bucket is missing and there is a gap in the data.
- **Fix:** rebuild the bucket from the closed 1m candles since the bucket start, plus the live tick. `SymbolCandleAggregator.Seed()` in `BotShared` already does this and can be reused per symbol.

2. The monthly timeframe ("1M") is handled as 1 minute

`Worker.cs:295` compares with `OrdinalIgnoreCase`, so "1M" matches "1m". The current 1m candle is appended to the monthly series as an extra bar. Only affects symbols with 60+ months of history (BTC, ETH and similar). `SymbolCandleAggregator.IsOneMinute` in `BotShared` has the same bug. **Fix:** use an ordinal (case-sensitive) comparison.

3. Money-flow lines can switch sides and push the probability the wrong way

In `McbProbabilityEngine.cs:753-764` (`AddMoneyFlowOnWavePivot`), take a **bearish** wave pivot where money flow went **up** (money flow does not confirm the bear setup):

- It gets `isBear=false` and is scored with `LiveBullScoreSplit`, using the tops' high prices.
- That gives a bull score up to about 0.99. It raises `liveBullDivScore`, which increases `LiveBullPct` and lowers the bear side through `oppBullPenalty`.
- The same happens the other way round for bull pivots.
- **Fix:** keep the money-flow line on the wave's side, and give it score 0 (or skip it) when money flow doesn't confirm.

Design problems

4. Ticks pile up in the queue

Every `LIVE_CANDLE_UPDATE` is handled one at a time (`prefetch=1`). Each one runs 10 queries of 2,500 candles and then writes every box.

- Bybit sends 1m candle updates about every second per symbol. With several symbols the queue grows and predictions fall behind.
- `_live1mCandles` is written but never read, and the `_InProgress` skip never triggers.

- **Fix:** acknowledge the message right away, store only the latest tick per symbol, and process it in a separate loop.

5. Box writes every tick

In `Worker.cs:583-793 (SaveBoxDivergenceCandidatesAsync)`:

- **No transaction:** the DELETE, the inserts and the box UPDATES run separately. The API can briefly read no candidates, or a half-updated set of boxes.
- **Every box is written on every tick**, even when nothing changed. Heavy database writes, and TradingBot trades on `position_type` while it keeps changing. Only update when the value differs (`WHERE position_type IS DISTINCT FROM @positionType`).
- **Manual or generated box directions get overwritten:** `boxes.position_type` is replaced for every box of the symbol, including boxes that BoxCreator or a user set. Is that intended?
- **Box timeframe is ignored:** `box.Timeframe` is read but not used, so a 1d box is switched on or off by 1h/4h/1d lines alike.

Smaller issues

- **Too little history:** 60 candles is the minimum, but money flow is a 60-period average and the std-dev uses 50 bars. With 60-200 candles (1w, young coins) the early values are distorted. 200+ is safer.
- **Lines carry bar positions, not times:** the frontend guesses the matching candle by comparing prices with candle wicks (`findBotLastBar` in `mcb-prediction-lines.ts`). Because of bug 1 those wicks don't match. Adding `AnchorTime/CurrentTime` to each line removes the guesswork.
- **0% money-flow lines are sent to the UI:** `AddMoneyFlowOnWavePivot` doesn't check `score > 0`.
- **Inconsistent filter:** historical money-flow divergences have no dominance filter, but wave divergences do.
- **Wrong log value:** the LIVE-TICK-STALE log puts `tick1m.Interval` where the symbol should be (`Worker.cs:342`).
- **Wrong comment:** says "Minimum 5 candles" but the code checks 10.
- **Retries and heartbeat:** retries are nested (up to 9 attempts), `Task.Delay` doesn't take the cancellation token, and there is a heartbeat database write on every tick.